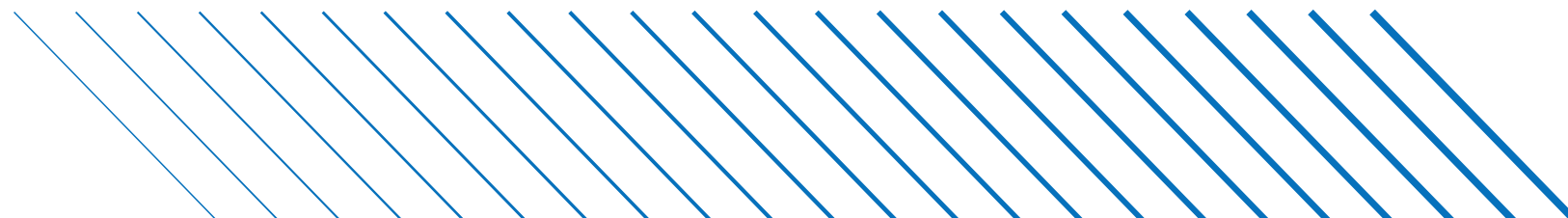
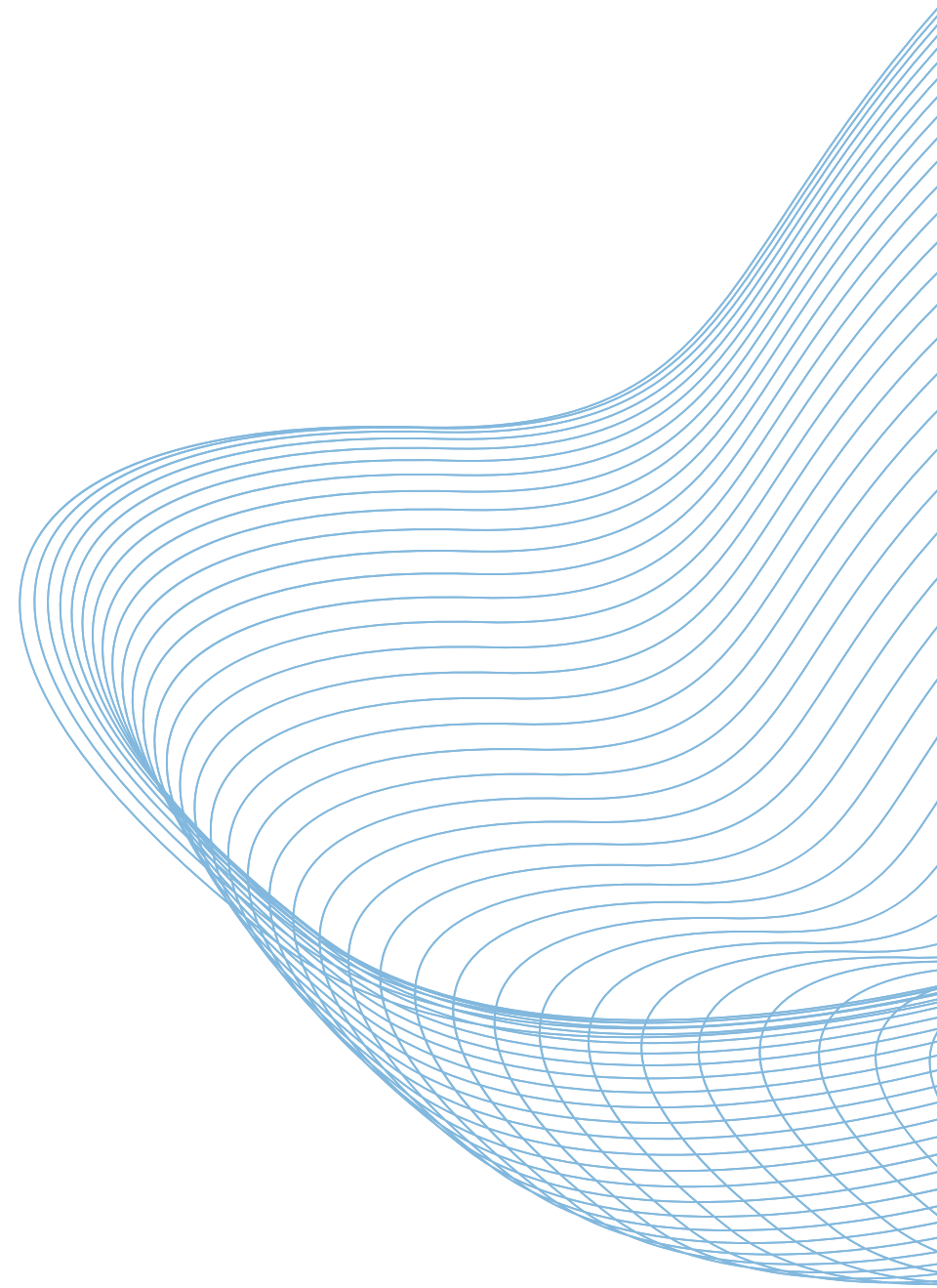


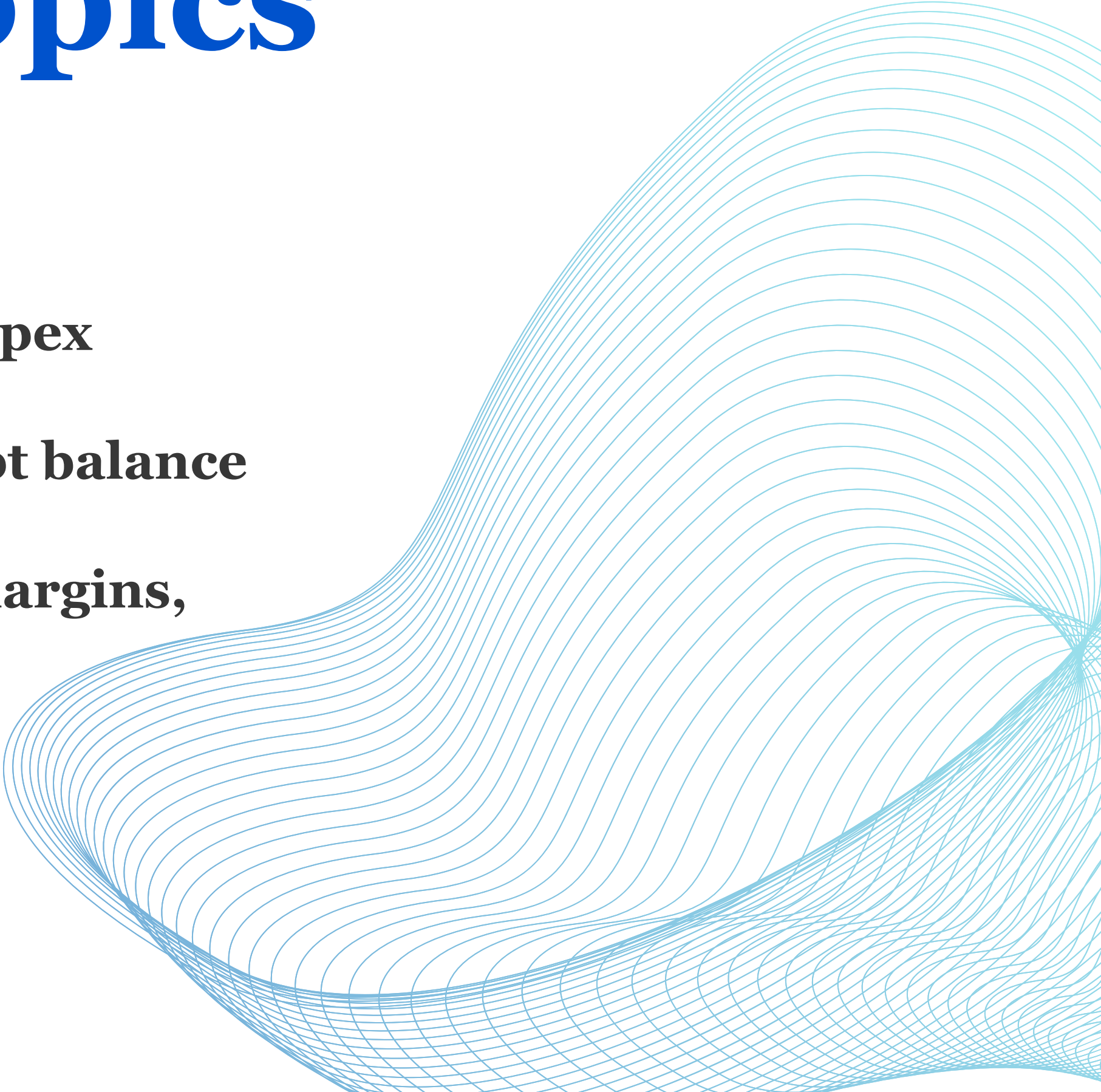
SageArk

Financial Modeling Mastery

Module 4: Three-Statement Projections



Overview of Topics

- **Linking the statements: IS, BS, CFS**
 - **Key projections: revenue, margins, capex**
 - **Building drivers for cash flow and debt balance**
 - **Validating assumptions for growth, margins,
and capital**
- 
- A decorative graphic consisting of numerous thin, light blue lines that flow and curve across the bottom right portion of the slide, creating a sense of movement and depth.



Projecting Revenue in Detail

Start with revenue assumptions

Step 1: Projecting Revenue

- **Simple growth rate (default)**
- **Units sold $\times$ price per unit**
- **Market share $\times$ market size**
- **Choose based on industry and data availability**

Example: High-Growth Recruiting Firm

- **Revenue linked to sales reps and productivity**
- **Use growth rates based on staff expansion**
- **Adjust based on historical trends**

Example: Mature Airline Company

- **Revenue linked to seat kilometers**
- **Use load factor and ticket price**
- **Calculate Revenue Seat Kilometers**

Example: Expansion-Stage Retailer

- **Link to store count and revenue per store**
- **Scale by location-specific factors**
- **Group by regions for simplicity**

Step 2: Projecting Expenses and Margins

- **COGS and Operating Expenses**
- **Link to revenue or drivers**
- **Link SG&A, R&D, S&M to revenue or units sold**

Example: Expense Projections for Frank Recruitment

- **Link Payroll to total staff**
- **Use average salary for forecast**
- **Keep growth assumptions reasonable**

Step 3: Projecting Operational Balance Sheet Items

- **AR, Inventory, AP, etc.**
- **Link these to revenue or operating expenses**
- **Use historical trends and growth assumptions**



Projecting Operating Expenses

Focus on SG&A, R&D, and marketing

Projecting Operating Expenses

- **SG&A: Link to revenue or headcount**
- **R&D: Link to product launch schedule**
- **Marketing: Link to sales growth or revenue**
- **Common assumption: % of revenue**

Example: Projecting R&D for Tech Firm

- **Link R&D to revenue growth**
- **Use historical R&D spend as baseline**
- **Adjust for future product launches**

Example: Marketing Expenses for Retailer

- **Link Marketing to revenue or growth rate**
- **Use historical spending as base**
- **Adjust for new campaigns**

Step 4: Projecting Non-Operating Balance Sheet Items

- **Debt, Equity, and Investments**
- **Debt: Interest-bearing liabilities and changes**
- **Equity: Retained earnings, stock issuance, buybacks**

Projecting Debt and Debt Repayments

- **Debt issuance: Cash inflow in CFF**
- **Debt repayments: Cash outflow in CFF**
- **Interest expense: Linked to average debt balance**

Projecting Equity

- **Equity issuances: CFF inflow**
- **Stock buybacks: CFF outflow**
- **Dividends: Non-cash tax adjustment in CFS**

Step 5: Projecting the Cash Flow Statement

- **CFO: Net Income + non-cash adjustments**
- **CFI: CapEx, investments**
- **CFF: Debt, equity, dividends**

Example: Cash Flow Projections for Tech Firm

- **CFO: Net income + depreciation + changes in working capital**
- **CFI: Investment in PP&E (CapEx)**
- **CFF: Financing from debt issuance, equity issuance**

Final Integration: Linking All Statements

- **IS to CFS: Link Net Income, adjustments**
- **CFS to BS: Link cash changes**
- **BS to IS: Link interest-bearing debt and equity balances**



Projecting Working Capital

Working capital changes and cash flow impacts

Projecting Working Capital: AR, AP, Inventory

- **AR: Link to revenue growth**
- **Inventory: Link to COGS or sales volume**
- **AP: Link to COGS or purchases**
- **Use historical data for working capital turns**

Projecting Other Assets and Liabilities

- **Prepaid Expenses:** Link to operating expenses
- **Deferred Revenue:** Link to sales or deferred contracts
- **Deferred Taxes:** Use effective tax rate
- **Other long-term liabilities:** Project based on known liabilities

Projecting CapEx & Depreciation

- **CapEx: Linked to growth or asset replacement**
- **Depreciation: Linked to CapEx and asset lives**
- **Link CapEx as a percentage of revenue or based on growth**

Projecting Debt Issuances & Repayments

- **Debt Issuance:** Linked to capital needs or financing strategy
- **Repayments:** Linked to maturity schedule
- **Interest:** Link to debt balance



Equity Issuances and Buybacks

- **Equity Issuances:** Linked to funding needs or growth plans
- **Stock Buybacks:** Linked to excess cash or strategic decisions
- **Dividends:** Linked to earnings payout ratio

Step 6: Projecting the Cash Flow Statement (CFS)

- **CFO: Start with Net Income, adjust for non-cash**
- **CFI: Account for CapEx, investments**
- **CFF: Include debt issuances and equity movements**

Example: CFS for Tech Firm

- **CFO:** Add back depreciation, adjust for working capital
- **CFI:** Track CapEx, investments
- **CFF:** Reflect debt issuance or buybacks

Integrating the 3 Statements

- **IS → Net Income → linked to CFS**
- **CFS → links to Balance Sheet (Cash)**
- **Balance Sheet → influences IS via debt/equity**

Final Integration: Linking All Statements

- **IS $\rightarrow$ CFS $\rightarrow$ BS**
- **Enforce balance with equity changes**
- **Stress-test assumptions in a flexible model**



Linking Debt and Equity Projections

Debt issuance, equity changes, and dividends

Projecting Debt Issuances

- **Debt Issuance: CFF inflow, increase debt in BS**
- **Debt Repayment: CFF outflow, decrease debt in BS**
- **Interest: Link to debt balance, expense on IS**

Projecting Equity Issuances & Buybacks

- **Equity Issuance:** CFF inflow, new shares issued
- **Stock Buybacks:** CFF outflow, reduce shares outstanding
- **Dividends:** Non-cash tax adjustment, linked to payout ratio

Step 7: Projecting the Cash Flow Statement (CFS)

- **CFO: Net Income + non-cash adjustments**
- **CFI: CapEx, acquisitions, investments**
- **CFF: Debt issuances, equity transactions, dividends**

Example: CFS for Tech Firm

- **CFO:** Add back depreciation, adjust for working capital
- **CFI:** Track CapEx, investments
- **CFF:** Reflect debt issuance or buybacks

Final Integration: Linking All Statements

- **IS $\rightarrow$ CFS $\rightarrow$ BS**
- **Enforce balance with equity changes**
- **Stress-test assumptions in a flexible model**

Key Takeaways

- **Understand the logic behind each statement**
- **Linking: Connect the IS, CFS, and BS seamlessly**
- **Focus on cash flow drivers**
- **Keep assumptions grounded in reality**

The End